PES UNIVERSITY

Department of Computer Science and Engineering

PROJECT SYNOPSIS / PROPOSAL

Course: UE24CS341A - Software Engineering | Semester: V

Project Title	Web Based ML Model Evaluator
Group Number	Team 13
Development Methodology	Agile (Scrum / Iterative Lifecycle)

1. Team Profile and Functional Ownership

The project is divided into complete functional areas so that each team member has clear ownership of one part of the system. All members will also participate in common project activities such as integration, reviews and testing.

Member	USN / SRN	Owned Functional Feature	Secondary Duties
Atharva Anand	PES1UG24CS092 PES1202401515	**User Authentication and Workspace Management** Registration, JWT login, RBAC and evaluation history.	Django authentication views and PostgreSQL user schema.
Atharva Rakesh Bidre	PES1UG24CS095 PES1202400624	**Dataset Ingestion and Schema Pre-Validation** Upload of model/dataset files, format checks, size checks, schema compatibility and target detection.	React upload interface and media storage setup.
Ayush M Shetty	PES1UG24CS103 PES1202403695	**Sandboxed Model Execution and Metrics Engine** Isolated model execution, task routing and metric calculation.	Celery task queue and Docker sandbox policy.
Bhavesh Velluru	PES1UG24CS115 PES1202400414	**Model Benchmarking, Visual Analytics and Report Export** Comparison tables, charts and PDF/JSON report generation.	Recharts interface and ReportLab PDF generation.

2. Problem Statement

Evaluating trained machine learning models for different datasets or model configurations often involves writing separate Python scripts or notebook code for loading the model, preparing the data, calculating metrics and creating plots. This becomes repetitive when several models need to be compared. Existing experiment tracking tools can also require changes to the training pipeline or additional SDK integration.

The proposed system provides a web interface where a user can upload a trained model and a test CSV dataset. The system validates the files, runs the evaluation in an isolated environment, displays the results and allows the user to compare models and export the results as reports.

3. Target Users / Stakeholders

- **ML Engineering Students and Researchers:** users who need to evaluate already-trained models without repeatedly writing evaluation scripts.
- **Course Evaluators and Teaching Assistants:** users who may need to evaluate model submissions using a common evaluation process.

4. Objectives

- Provide a web-based interface for uploading trained models and test datasets.
- Validate file format, size and feature compatibility before model execution.

- Run model inference inside an isolated Docker environment.
- Automatically identify classification or regression evaluation and calculate the relevant metrics.
- Display useful evaluation plots and metric summaries.
- Allow multiple evaluated models to be compared on the same dataset.
- Provide downloadable PDF and JSON summaries of evaluation results.

5. Product Scope

In scope: classical machine learning models supported by scikit-learn and ONNX models, evaluated using tabular CSV datasets. The system covers model/dataset upload, validation, sandboxed execution, metric calculation, visualization, comparison and report export.

Out of scope for Version 1: model training, real-time streaming inference APIs, and unstructured media pipelines such as image, video and NLP data.

6. Functional Features

- User account registration, JWT authentication and workspace evaluation history.
- Model and dataset upload with format, size and schema validation.
- Sandboxed model execution with automatic classification/regression task detection.
- Metric calculation for classification and regression tasks.
- Interactive visual dashboard for evaluation results.
- Side-by-side benchmarking of multiple evaluated models.
- PDF and JSON summary report export.

7. Plan of Work

Sprint	Timeline	Planned Work / Evidence	Owner
1	Weeks 1–2	Project setup, Django REST backend, JWT authentication APIs and initial Jira backlog.	Atharva A.
2	Weeks 3–4	File ingestion, CSV schema validation and upload interface.	Atharva B.
3	Weeks 5–6	Docker sandbox worker, task execution and ML metric services.	Ayush S.
4	Weeks 7–8	Model comparison and ReportLab-based report generation.	Bhavesh V.
5	Weeks 9–10	React dashboard integration, charts and SonarQube review.	All members
6	Weeks 11–12	GitHub Actions CI/CD, security hardening, system demonstration and final report preparation.	All members

8. Proposed Technology and Tools

- **Frontend:** React.js (Vite), Tailwind CSS, Recharts / Plotly.js
- **Backend:** Python 3.11, Django, Django REST Framework, Celery and Redis
- **ML / Reports:** scikit-learn, NumPy, Pandas, ONNX Runtime and ReportLab
- **Development / Testing:** Docker, PostgreSQL 15, Git/GitHub, GitHub Actions, SonarQube and Postman